

Fig. 4: NAND gate as inverter

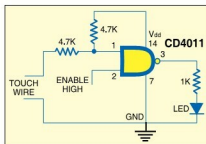


Fig. 5: NAND gate as switch

logic '1' to the gate input. As a result, the bulb will not glow. When any of the doors is not properly shut, logic '0' is applied to the gate. The bulb glows to indicate that any of the doors is not properly shut.

Inverter. In the inverter circuit shown in Fig. 4, when you give input '1', the output is '0'. When the input is '0', you will receive the output as '1.'

Switch. The switch circuit in Fig. 5 gives a high output when you touch both the wires together. Working of the circuit is very simple. As pin 2 is connected to the power supply, it always has a high input. When the wires are not touched, the circuit is open and pin 1 is high. When the wires are touched, you will get logic '0' at pin 1. As a result, the LED will glow.

LED flasher circuit. This is a small circuit used to flash two LEDs after specific interval of time with the help of a NAND gate. Here CD4011 is wired in astable multivibrator mode, generating 50 per cent duty cycle square wave at approx. 1Hz frequency. This circuit can be used as the eye of a toy or as an indicator.

Working of the circuit is very simple. Here all the four gates are used. When you switch on the power supply, the voltage at pins 1 and 2 is high. As you can see from the truth table, output pin 3 becomes low, which is supplied to pins 5 and 6. From the truth

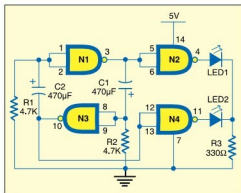


Fig. 6: LED flasher circuit

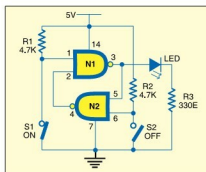


Fig. 7: Toggle switch circuit

table, if both the inputs are low, you will get a high output. As a result, LED1 connected to pin 4 starts glowing. Time period for which the LED glows depends on value of the capacitor used. If you increase value of the capacitor, charging time will increase and LED1 will glow for a longer period. If you decrease the value, glowing time will also decrease.

As the capacitor connected to pin 3 starts charging, pins 8 and 9 will receive a high pulse. As a result, pin 10 becomes low and the output from pin 11 makes LED2 glow. Capacitor C2 discharges on pins 1 and 2, which again become high and the whole cycle repeats.

Note that you can also connect a relay in place of the LED so that after certain period of time, one device turns on while the other turns off.

Toggle switch. This circuit lets you switch on and off a device with the touch of your hand. The benefit of using a touch sensor is that you will not get any electrical shock. You can use it to switch on/off your night lamp

or motor by simply connecting a relay at the output.

The circuit is based on NAND gate CD4011. Out of the four gates of CD4011 only two are used here.

Working of the circuit is very simple. In standby mode, both the switches are closed and the current flows through resistors R1 and R2 to ground. When you touch the 'on' switch, pin 1 receives logic zero. Now whatever be the input at pin 2, you will get a high output at pin 3 (refer the truth table). At this time, the input at pin 6 is high as 'off' switch is not pressed. Pin 5 is also high as the output from pin 3 is fed back to pin 5. As a result, pin 4 goes low and the LED connected to it starts glowing.

When you again touch the 'off' switch, the reverse action will take place. That means when you press the switch, pin 6 will receive logic '0' at its input and whatever be the input at pin 5, you will receive a high at pin 4. At this time, pins 1 and 2 both are high. Therefore you will receive logic '0' at pin 3. As a result, the LED connected at the output will go off.

In place of the LED, you can also connect a relay with the help of a transistor. In this case, you can use a BC547 transistor and supply voltage according to the relay voltage rating. As the working voltage of the IC varies from 5V to 15V, do not supply or use relay of more than 16V. Otherwise, it will damage your IC.

If you want to make a touch sensor at your home, take a conducting plate (may be copper) and glue it to some non-conducting material like wood or plastic in close proximity to each other. As the human body is also an electrical conductor, touching the surface of the conductor results in a distortion of the screen's electrostatic field, measurable as a change in capacitance. You can also use a ready-made sensor available in the market or a push-to-on switch. ●

The author is a B.Tech